

Shaping the nuclear future by the accounting valuation of existing nuclear electricity costs

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Despite the liberalization that was supposed to establish the market price mechanism as the stickyard for valuing electricity, the full cost of nuclear electricity has remained a central reference in French energy debates and in the relationship between the State and EDF. While market prices fluctuate significantly, this reference serves as a benchmark for ensuring a “fair remuneration” for the company and its historical assets. However, numerous controversies and tensions have punctuated this cost calculation since liberalization, reflecting profound divergences—not about the past, but about ambitions for the future. These tensions concern, first, the future costs of decommissioning and waste management, but more significantly, the costs associated with the extension and renewal of the nuclear fleet. In this way, considerations about the future of nuclear energy have influenced the valuation of historical nuclear production, shaping debates about how the value of electricity should be calculated.

This article aligns with prior work on valuation, particularly the valuation of electricity through purchase tariffs (Pallesen, 2016), market prices (Reverdy & Breslau, 2019), the valuation of resources mobilized for production (Labussière & Nadaï, 2018), or nuclear waste (Saraç-Lesavre, 2021). These studies have highlighted the assumptions, reasoning, and calculation methods underpinning economic valuation, reflecting various politico-institutional framings (Kjellberg & Helgesson, 2010; Fourcade, 2011), economic logics (Silvast, 2017), administrative rationales (Asdal, 2014), and collective mobilizations (Dubuisson-Quellier, 2013). Implicitly, these works also reveal the more immediate stakes of valuation processes, including investment decisions, resource allocation, or the regulation of market exchanges. Valuation practices typically produce economic effects whose anticipation and assessment are embedded in the markets (Reverdy & Breslau, 2019). In particular, when valuation processes are established or modified through political decisions or economic regulations, they must account for economic effects and their potential political consequences, such as questions of justice, impacts on economies, and shifts in power dynamics (Vogel, 2018).

This article argues that the accounting calculation of nuclear electricity’s value represents a battleground for competing visions regarding the status of historical nuclear production, the status of EDF as a company, its strategic and investment capacities, and the financing of future nuclear projects. This accounting work crystallizes struggles between EDF and successive governments, especially as its outcomes are used as a reference point for evaluating EDF’s revenues, whether derived from market mechanisms or public tariffs.

Our research draws on interviews with stakeholders (EDF, CRE, CEA, DGEC, ministerial offices), archival work, and “embodied archives” in the sense defined by Gildas Loirand (Loirand, 2021), gathered during three years of professional experience within those actors’ networks.

Case study

Throughout this history, a tension emerges between two financing approaches. The first

envisioned EDF as an autonomous company capable of financing its development and investments according to market logics, a perspective advocated by EDF through direct borrowing on financial markets (Bergougnoux, 1987). This vision shapes EDF's leadership strategy today, seeking autonomy via greater market exposure. The second, promoted by parts of the administration, evaluates production costs historically, enabling the State to limit EDF's remuneration to operating costs and depreciation of past investments, capturing the economic rent to serve its policies. This isolates the past from the future, allowing the State to allocate resources to technologies of its choice and open the sector to new actors.

This tension crystallized in debates among EDF, the Court of Auditors, and the government over the valuation of EDF's nuclear electricity. These debates have shaped EDF's remuneration and politically driven market regulations. Three periods are discernible:

Before liberalization, EDF engineers linked load demands and investment decisions to social return maximization (Yon, 2014). Known as "long-term marginal cost pricing," this approach followed cost-recovery logic, where rent serviced debt, ensured capital reproduction, and supported expansion aligned with consumption projections.

After liberalization, enthusiasm grew for market opportunities, including at EDF (Reverdy, 2014). However, market-based valuation, rooted in neoclassical economics (Doganova, 2023), soon revealed political challenges like the "nuclear rent." Until 2007, tariffs for end consumers persisted, calculated using pre-liberalization methods. From 2006, political efforts to redistribute the "nuclear rent" underscored cost evaluation issues for ensuring "fair remuneration" for EDF and financing projects like Flamanville.

In 2010, the NOME law introduced ARENH, a new mechanism performing politically determined rent sharing. As costs rose with aging infrastructure, EDF sought to influence or abolish ARENH to rebuild investment capacity (Creti & Fontini, 2019). EDF proposed an economic approach ensuring "remuneration consistent with business risk" while enabling asset renewal. Conversely, the government's full-cost accounting constrained EDF's autonomy, allowing nuclear rent redistribution to consumers and discreet electricity taxation for renewable investments.

EDF's demands continue its historical calls for autonomy. Meanwhile, the State, particularly the DGEC, advocates for separating historical nuclear assets (including Flamanville 3) from new projects. Full-cost accounting underpins the State's control of EDF, considering market pressures insufficient (given EDF's dominant position) or irrelevant (as market prices are deemed unsuitable for electricity).

This divide is reflected in the very calculation mechanisms used to determine the remuneration of fissile capital (Pottin, 2024), which has proven to be poorly suited to the current market environment of late capitalism (financialized, liquid, etc.). Stabilized calculation frameworks embed normative regulatory assumptions. EDF's weighted average cost of capital reflects greater market exposure, while CRE's government-requested calculations secure a revenue corridor, reducing market volatility. These approaches diverge by 5 euros per MWh, a discrepancy exceeding €1 billion annually.

Beyond EDF-State tensions, the industrial choice of a standardized fleet poses systemic risks for aging infrastructure, more prone to technical failures at fleet scale. Recognized institutionally, "generic incidents" capture these risks in valuation formulae, with discrepancies around 2.7 euros per MWh between CRE and EDF.

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